

Sequences and Basic Series Convergence Tests

ANSWER KEY



Convergence of sequences

- 1 Determine whether the sequence $a_n = \frac{3n+1}{n}$ converges, and if so, find its limit.
 $a_n = 3 + \frac{1}{n} \rightarrow 3$ as $n \rightarrow \infty$. Converges to 3.
- 2 Determine whether the sequence $a_n = (-1)^n$ converges.
The sequence oscillates between -1 and 1 and never settles on a single value, so it diverges.

Geometric series

- 3 Determine whether $\sum_{n=1}^{\infty} 3\left(\frac{1}{2}\right)^n$ converges, and if so, find its sum.
Geometric series with $r = \frac{1}{2}$, $|r| < 1$: converges. Sum $= \frac{a}{1-r} = \frac{3}{1-1/2} = 6$.
- 4 Determine whether $\sum_{n=1}^{\infty} 2^n$ converges.
Geometric with $r = 2$, $|r| \geq 1$: diverges.

The n th-term (divergence) test

- 5 Use the n th-term test to show $\sum_{n=1}^{\infty} \frac{n}{n+1}$ diverges.
 $\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \neq 0$. Since the terms don't approach 0, the series diverges by the n th-term test.

The integral test

- 6 Use the Integral Test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.
Let $f(x) = \frac{1}{x^2}$, positive, continuous, decreasing for $x \geq 1$. $\int_1^{\infty} \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_1^{\infty} = 0 - (-1) = 1$, a finite value.
By the Integral Test, the series converges.
- 7 Use the Integral Test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges (the harmonic series).
 $\int_1^{\infty} \frac{1}{x} dx = [\ln x]_1^{\infty} = \infty$. Diverges. So the harmonic series diverges by the Integral Test.

The comparison and limit comparison tests

- 8 Use the Direct Comparison Test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges, comparing to $\sum_{n=1}^{\infty} \frac{1}{n^2}$.
 $\frac{1}{n^2+1} < \frac{1}{n^2}$ for all $n \geq 1$, and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges (p-series with $p = 2 > 1$). By Direct Comparison, $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges.

- 9 Use the Limit Comparison Test to determine whether $\sum_{n=1}^{\infty} \frac{2n}{n^2+3}$ converges, comparing to $\sum \frac{1}{n}$.
 $\lim_{n \rightarrow \infty} \frac{\frac{2n}{n^2+3}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{2n^2}{n^2+3} = 2$, a finite positive value. Since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges (harmonic), $\sum \frac{2n}{n^2+3}$ also diverges by the Limit Comparison Test.

Visualizing sequence convergence

- 10 Plot the first 6 terms of $a_n = \frac{3n+1}{n}$ (from the earlier question) and describe how the points behave as n increases.
 $a_1 = 4, a_2 = 3.5, a_3 \approx 3.33, a_4 = 3.25, a_5 = 3.2, a_6 \approx 3.17$. The points decrease and cluster tighter and tighter around the horizontal line $y = 3$, visually confirming the limit found algebraically.

Monotonic and bounded sequences

- 11 Determine whether $a_n = \frac{n}{n+1}$ is monotonic (always increasing or always decreasing), and find an upper bound.
 $a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)} > 0$, so the sequence is strictly increasing. Since $a_n = 1 - \frac{1}{n+1} < 1$ for all n , it is bounded above by 1.
- 12 State the Monotonic Sequence Theorem, and use it to conclude that $a_n = \frac{n}{n+1}$ converges.
The Monotonic Sequence Theorem states that every bounded, monotonic sequence converges. Since $a_n = \frac{n}{n+1}$ is increasing and bounded above by 1 (shown above), it must converge (in fact to 1).

The p-series test

- 13 State the p-series test for $\sum_{n=1}^{\infty} \frac{1}{n^p}$, and classify convergence for $p = 1, p = \frac{3}{2}$, and $p = \frac{1}{2}$.
 $\sum \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$. $p = 1$: diverges (harmonic). $p = \frac{3}{2} > 1$: converges. $p = \frac{1}{2} \leq 1$: diverges.
- 14 Use the p-series test (after simplifying) to determine whether $\sum_{n=1}^{\infty} \frac{5}{n^{0.9}}$ converges.
This is $5 \sum_{n=1}^{\infty} \frac{1}{n^{0.9}}$, a constant multiple of a p-series with $p = 0.9 \leq 1$. Diverges.

Series review: choosing the right test

- 15 For each series, name the most efficient test to apply (do not fully evaluate):
- | | |
|---|---|
| a $\sum \frac{1}{5^n}$ Geometric series test ($r = \frac{1}{5}$) | b $\sum \frac{1}{n^3}$ p-series test ($p = 3$) |
| c $\sum \frac{n+1}{2n+3}$ nth-term test (terms $\rightarrow \frac{1}{2} \neq 0$) | d $\sum \frac{1}{n^2+2n}$ Direct/Limit Comparison to $\sum \frac{1}{n^2}$ |

Sequences defined recursively

- 16** A sequence is defined recursively by $a_1 = 2$, $a_{n+1} = \frac{a_n + 6}{2}$. Compute a_2 , a_3 , and a_4 , and conjecture the limit if the sequence converges.
 $a_2 = \frac{2+6}{2} = 4$. $a_3 = \frac{4+6}{2} = 5$. $a_4 = \frac{5+6}{2} = 5.5$. The terms appear to approach 6 (if $L = \frac{L+6}{2}$, then $2L = L + 6 \Rightarrow L = 6$).
- 17** Verify that if a recursively defined sequence $a_{n+1} = \frac{a_n + 6}{2}$ converges to a limit L , then L must satisfy $L = 6$.
If $a_n \rightarrow L$, then $a_{n+1} \rightarrow L$ as well, so taking the limit of both sides: $L = \frac{L+6}{2} \Rightarrow 2L = L + 6 \Rightarrow L = 6$, confirming the conjecture.
-

Geometric series with algebraic manipulation

- 18** Find the sum of $\sum_{n=2}^{\infty} \left(\frac{2}{3}\right)^n$ (note the series starts at $n = 2$, not $n = 0$).
Using $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ and subtracting the first two terms ($n = 0, 1$): $\sum_{n=2}^{\infty} \left(\frac{2}{3}\right)^n = \frac{1}{1-2/3} - 1 - \frac{2}{3} = 3 - \frac{5}{3} = \frac{4}{3}$.
- 19** Express the repeating decimal $0.\overline{45}$ as a geometric series and find its exact fractional value.
 $0.454545... = \sum_{n=1}^{\infty} 45\left(\frac{1}{100}\right)^n = \frac{45/100}{1-1/100} = \frac{45/100}{99/100} = \frac{45}{99} = \frac{5}{11}$.
-

Telescoping series

- 20** Evaluate the telescoping series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right)$ by examining its partial sums.
 $S_N = \sum_{n=1}^N \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{N+1}$ (all middle terms cancel). As $N \rightarrow \infty$: $S_N \rightarrow 1 - 0 = 1$. The series converges to 1.